Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2024

Bachelor of Computer Applications SEMESTER - I

Course Code: BCA101-1 **Max Marks: 100**

Course Name: FOUNDATIONAL MATHEMATICS **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Reformulate statements from common language to formal logic using the rules of propositional and predicate calculus and assess the validity of arguments.

CO2: formulate and interpret statements presented and determine their validity by applying the rules and methods of propositional logic.

CO3: construct elementary proofs using ordinary and strong induction in the context of studying the properties of recursion.

CO4: apply basic counting principles including the pigeonhole principle and rules for counting permutations and combinations.

CO5: apply the logical structure of proofs and work symbolically with connectives and quantifiers to produce logically valid, correct and clear arguments

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Section A **3*8=24**

Answer ALL the questions:

1)	Show that conditional statement "$\neg(p \to q) \to \neg q$" is a tautology by using truth tables.	**3**	(CO1, L1)
2) a)	Construct a truth table for compound proposition $(p \to q) \wedge (\neg p \to r)$.	**3**	(CO1, L1)
	OR		
b)	State and prove the Modus Ponens law without using truth table.	**3**	(CO1, L1)
3)	Translate these statements into English, where $P(x)$ is "x is a rabbit" and $Q(x)$ is "x hops" and the domain consists of all animals: (i) $\forall x(P(x) \to Q(x))$, (ii) $\forall x(P(x) \to Q(x))$, (iii) $\exists x(P(x) \wedge Q(x))$.	**3**	(CO2, L1)
4)	Show that " I must have watched TV in the evenings " is a valid conclusion from the premises "If I study, I will not fail in the examination", "If I do not watch TV in the evenings, I will study" and "I failed in the examination".	**3**	(CO2, L1)
5) a)	Prove that n is an odd positive integer if and only if $5n + 6$ is odd.	**3**	(CO3, L3)
	OR		
b)	Show that if a and b are real numbers and $a \neq 0$, then there is a unique real number r such that $ar + b = 0$.	**3**	(CO3, L3)

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6)	If n is a positive integer, then construct a formula for $1+3+\cdots+(2n-1)$ and prove it by mathematical induction.	**3**	(CO3, L3)
7)	How many different license plates can be made if each plate contains a sequence of three uppercase English letters followed by three digits? Explain the mathematical principle used to get the answer.	**3**	(CO4, L3)
8) a)	Show that among any group of five (not necessarily consecutive) integers, there are two with the same remainder when divided by 4.	**3**	(CO4, L4)
	OR		
b)	How many permutations of the letters $ABCDEFGH$ contain the string ABC?	**3**	(CO4, L4)

Section B **8*7=56**

Answer ALL the questions:

9)	Develop the series of logical equivalences to show $(p \to q) \to (r \to s)$ and $(p \to r) \to (q \to s)$ are not logically equivalent.	**8**	(CO1, L1)
10) a)	When planning a party you want to know whom to invite. Among the people, you would like to invite three touchy friends. You know that if Jasmine attends, she will become unhappy if Samir is there, Samir will attend only if Kanti will be there, and Kanti will not attend unless Jasmine also does. Which combinations of these three friends can you invite so as not to make someone unhappy?	**8**	(CO2, L2)
	OR		
b)	Using truth table verify that the distributive law for disjunction over conjunction.	**8**	(CO2, L2)
11)	(i) Determine the truth value of each of these statements if the domain of each variable consists of all real numbers, and justify your answer: (a) $\exists x \forall y(y \neq 0 \to xy = 1)$, (b) $\forall x \exists y(x + y = 1)$, (c) $\exists x \exists y(x + y \neq y + x)$, (d) $\forall y \exists x(y \neq 0 \to xy = 1)$. (ii) Let us consider the statement $P(x, y) : x \leq y^2$. Find the truth value of $\exists x \forall y P(x, y)$ if the domain consists of (a) the positive real numbers (b) the integers (c) the nonzero real numbers. Justify your answer.	**8**	(CO5, L2)
12)	Test the validity of the conclusion $Q \to P$ from the premises $U \to R$, $(R \to S) \to (P \vee T)$, $\neg T$ and $Q \to (U \wedge S)$.	**8**	(CO5, L2)
13) a)	Prove by at least three different methods: "If $3n + 2$ is even, then n is even".	**8**	(CO3, L3)
	OR		
b)	Prove that if n is an integer, these four statements are equivalent: (i) n is even (ii) $n + 1$ is odd (iii) $3n + 1$ is odd (iv) $3n$ is even.	**8**	(CO3, L3)
14)	Prove by mathematical induction that, for all positive integers $n \geq 1$, $1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \cdots + \frac{1}{1+2+3+\cdots+n} = \frac{2n}{2n+1}$	**8**	(CO3, L2)
15) a)	How many positive integers between 5 and 31: (i) a) are divisible by 3? Which integers are these? (ii) are divisible by 4? Which integers are these? (iii) are divisible by 3 and by 4? Which integers are these?	**8**	(CO4, L3)
	OR		
b)	A bowl contains 10 red balls and 10 blue balls. A woman selects balls at random without looking at them: (i) How many balls must she select to be sure of having at least three balls of the same color? (ii) How many balls must she select to be sure of having at least three blue balls?	**8**	(CO4, L3)

Section C **10*2=20**

Answer ALL the questions:

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16)	Construct logical expressions from the statements “Someone in your school has visited US” and “Everyone your class has studied Java and C++” in four different ways by varying domain and by using predicates with one and with two variables.	**10**	(CO5, L2)
17) a)	Explain your answer for the following: (i) A box contains 2 white balls, 3 black balls and 4 red balls. In how many ways can 3 balls be drawn from the box, if at least one black ball is to be included in the draw? (ii) There are 5 boys and 5 girls. In how any ways they can be seated in a row so that all the girls do not sit together? (iii) (c) In a party every guest shakes hand with every other guest. If there was total of 105 handshakes in the party, find the number of persons present in the party?	**10**	(CO4, L3)
	OR		
b)	What is the coefficient of x^9 in $(2-x)^{19}$? What is the coefficient of $x^8.y^9$ in the expansion of $(3x+2y)^{17}$?	**10**	(CO4, L3)

Revised Bloom's Taxonomy (RBT) Levels		
L1 - Remember	L2 - Understand	L3 - Apply
L4 - Analyse	L5 - Evaluate	L6 - Create

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